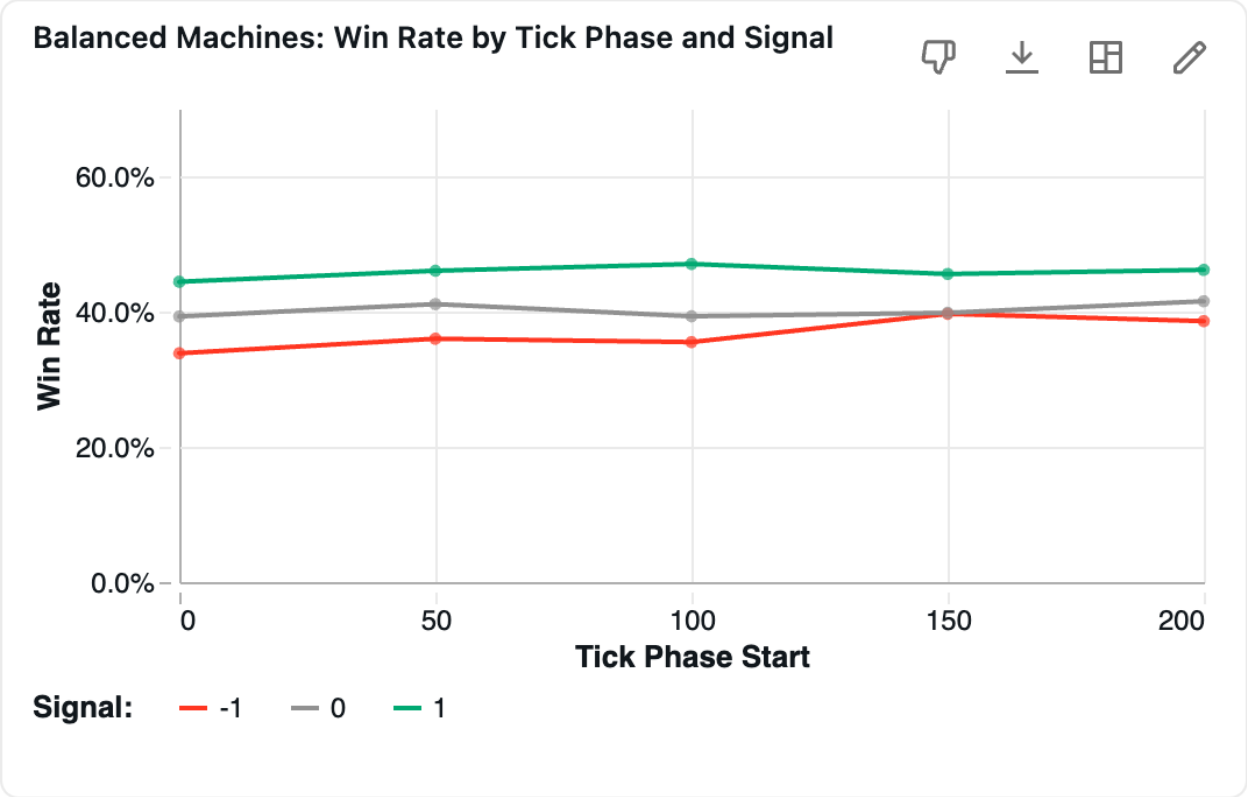
Drift Hazard & Drought Profile Analysis

Executive Summary

The analysis reveals **distinct drift patterns across machine types**, with **signal value** being the dominant predictor of win probability rather than episode progression (tick phase). **Steady machines show remarkable stability** (52-64% win rates), while **jackpot machines exhibit high volatility** (16-31% win rates) but no consistent decay pattern. Balanced machines fall in between (34-47% win rates) with signal-driven variations.

Key Findings by Machine Type

1. Balanced Machines: Signal-Driven Performance



Balanced machines show moderate win rates (34-47%) with signal as the key driver

• **Signal +1** (positive): Highest win rate, ranging from 44.4% (phase 0) to 47.0% (phase 2), with mean payout increasing from 1.17 to 1.35 by phase 4
• **Signal 0** (neutral): Mid-range win rate of 39-42%, relatively stable across phases
• **Signal -1** (negative): Lowest win rate of 34-40%, showing some recovery in later phases

No evidence of systematic drift/decay: Win rates fluctuate but don't consistently decline as episodes progress.

2. Jackpot Machines: High Variance, No Clear Drift

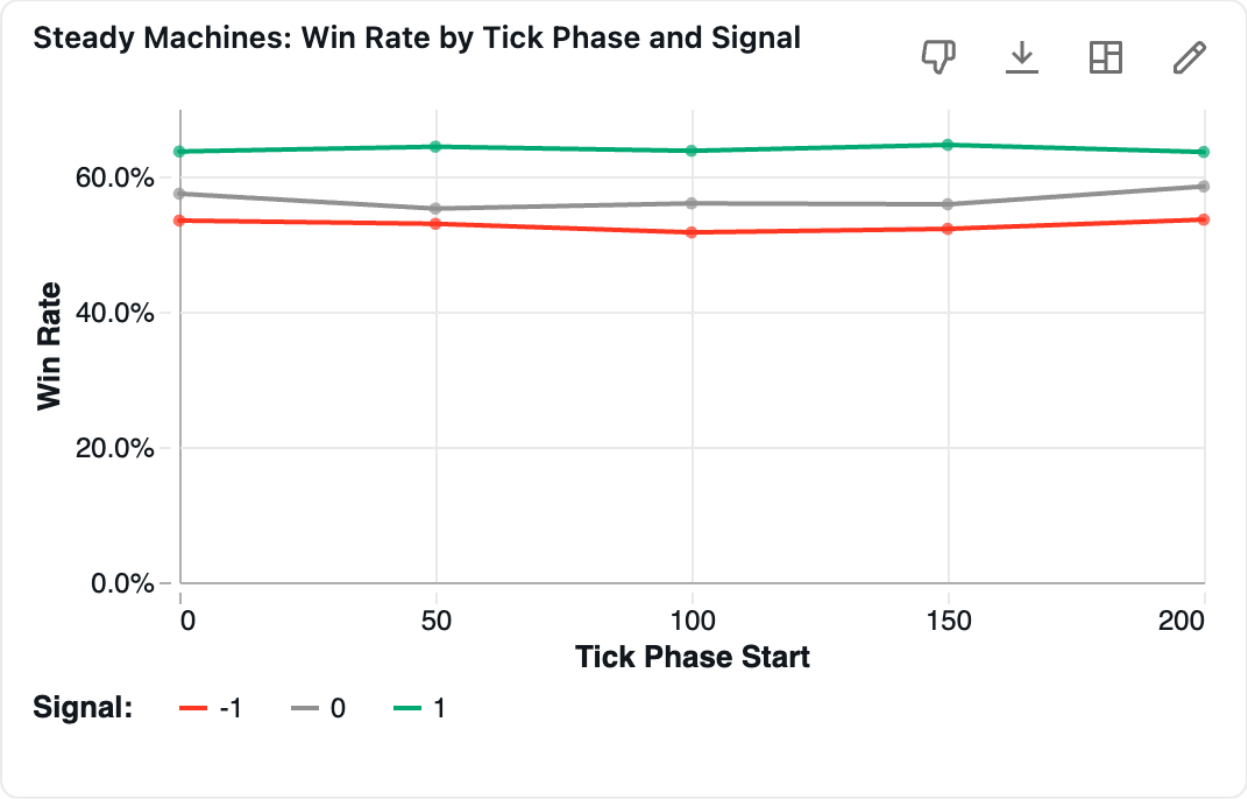


Jackpot machines have the lowest overall win rates (16-31%) but highest payouts when wins occur

• **Signal +1**: Win rate ranges 28-32%, with consistently high mean payout (~1.44-1.53) and jackpot rate (3.4-4.0%)
• **Signal 0**: Win rate shows a **U-shaped pattern** — starts at 20.8% (phase 0), drops to 17.8% (phase 2), then **surges to 27.5%** (phase 4)
• **Signal -1**: Lowest win rate (16-18%), but mean payout actually increases from 0.86 to 0.97 across phases

Jackpot rate remains elevated (2-4%) across all phases, indicating the high-risk/high-reward nature persists throughout episodes.

3. Steady Machines: Exceptional Stability



Steady machines live up to their name with remarkably consistent win rates (52-64%)

• **Signal +1**: Highest and most stable win rate (63.6-64.6%), with minimal variation across all phases
• **Signal 0**: Consistent 55-58% win rate throughout
• **Signal -1**: Lower but still stable 52-54% win rate

Jackpot rates are minimal (0-0.2%), confirming these machines prioritize frequent small wins over rare large payouts. Mean payouts range only 0.85-1.07.

Drift Hazard Assessment

Does Win Probability Decay with Episode Progression?

No systematic drift/decay detected. Across all three machine types:

1. **Win rates do NOT consistently decline** as tick phases progress from 0 to 200+
2. Some machines show **slight increases** in later phases (e.g., Jackpot Signal 0 jumps from 17.8% to 27.5%)
3. **Signal value is the dominant factor**, not relative episode progression (t/cutoff)

Signal as the Primary Predictor

The data strongly suggests that **signal value acts as a regime indicator**

• **Signal +1**: Consistently highest win rates across all machine types
• **Signal -1**: Consistently lowest win rates
• **Signal 0**: Middle ground

This pattern holds across all tick phases, indicating signal represents an **underlying state** of the machine rather than a transient feature.

Drought Profile Implications

The "drift" condition may not refer to temporal decay but rather to:

1. **Signal-based regime shifts**: Machines in negative signal states (-1) operate in a lower-probability regime
2. **Machine type characteristics**: Jackpot machines naturally have extended droughts (low win rates) regardless of phase
3. **Volatility patterns**: Jackpot machines show the most phase-to-phase variation, while steady machines maintain consistency

The absence of systematic decay suggests that **consecutive zero-payout streaks (droughts) are driven by the machine's current signal state** rather than accumulated play history or episode progression.